The Ankyrin repeat and SOCS box containing protein 4 (Asb-4) is highly expressed in the cytosol of hypothalamus cells during the well-fed state but is down-regulated during fasting. In addition, the protein called Insulin Receptor Substrate 4 (IRS4), known to be involved in appetite regulation, is expressed in the cytosol of the same cells as Asb-4. Accordingly, researchers hypothesized that Asb-4 suppresses appetite through interactions with IRS4.

Small peptides such as myc (EQKLISEEDL) and Flag (DYKDDDDK) are often attached to recombinant proteins to facilitate their purification and detection. To better understand interactions between Asb-4 and IRS4, HEK293 cells were transfected with plasmids encoding IRS4 and different forms of Asb-4, each tagged with either myc or Flag at their N-termini (Flag-IRS4 and myc-Asb-4, respectively). All transfections contained 1 μg of Flag-IRS4 plasmid as well as varying amounts of either myc-Asb-4 plasmid or a plasmid in which a domain called the sb domain of Asb-4 was deleted (myc-Asb-4Δsb). Flag-IRS4 levels from each set of cells were then quantified, as shown in Figure 1.

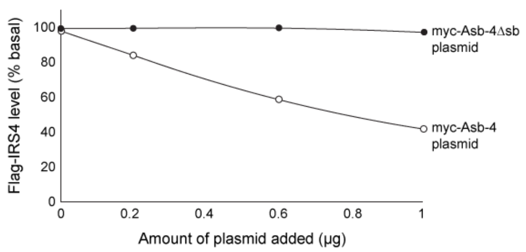


Figure 1 Flag-IRS4 levels in cells with varying amounts of myc-Asb-4 or myc-Asb-4Δsb plasmid added

In a follow-up experiment called coimmunoprecipitation, transfected cells were lysed and the lysates were incubated in columns containing beads linked to α-Flag, an antibody that binds Flag. Proteins that did not bind were washed away. The bound proteins were then visualized by western blot analysis using α-Flag, α-myc, and an antibody against ubiquitin (α-ub). The results are shown in Figure 2. The α-myc antibody detected the presence of myc, indicating that myc-Asb-4 binds to Flag-IRS4.

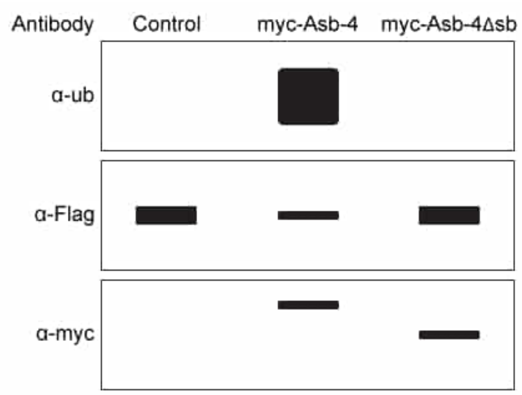


Figure 2 Ubiquitin, Flag-IRS4, and myc-Asb-4 levels in cells transfected with Flag-IRS4 and either an empty vector (control), myc-Asb-4 plasmid, or myc-Asb-4Δsb plasmid

Which of the following most accurately describes the interaction between the antibody α-Flag and its target, Flag-IRS4?

- ◯ A. A covalent protein-protein interaction [49%]

✔ ◉ B. A noncovalent protein-protein interaction [41%]

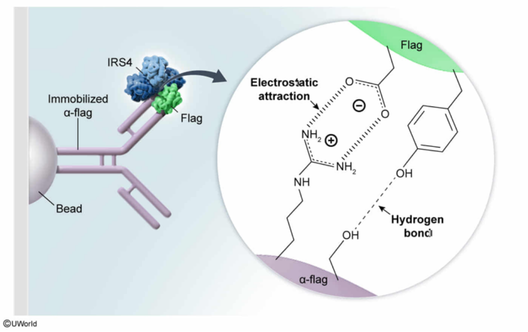
◯ C. A covalent protein-carbohydrate interaction [6%]

◯ D. A noncovalent protein-carbohydrate interaction [2%]

Correct
41% answered correctly

Explanation:

Antibodies participate in noncovalent protein-protein interactions



Antibodies **are proteins** that function in the immune system to recognize foreign particles known as **antigens**. Specifically, antibodies mark foreign antigens for destruction (eg, phagocytosis) by immune cells. Each antibody contains a variable region that binds to a specific chemical structure within an antigen, known as an **epitope**. The epitope is commonly a sequence of amino acids within a protein or peptide. Antibodies bind their epitopes through **noncovalent interactions** such as hydrogen bonding and electrostatic attractions.

The passage states that cell lysates were incubated with α-Flag, which is an antibody that binds the Flag peptide. Therefore, the Flag sequence was the epitope, and it was this portion of Flag-IRS4 to which the antibody noncovalently bound.

Because both the antibody (α-Flag) and its target (Flag-IRS4) are proteins, the interaction between them is a protein-protein interaction. Because antibodies generally bind their targets noncovalently, the interaction can be classified more specifically as a **noncovalent protein-protein interaction**.

(Choice A) Covalent interactions occur when atoms share electrons. *Within* a protein, covalent peptide bonds link amino acids to each other, but most interactions *between* proteins are noncovalent. Antibodies generally bind their epitopes through *noncovalent* interactions such as hydrogen bonds and electrostatic attractions.

(Choices C and D) Antibodies and Flag-IRS4 are both proteins, so they bind through protein-protein interactions rather than protein-carbohydrate interactions.

Educational objective:
Antibodies are proteins that noncovalently bind specific chemical groups on proteins known as epitopes. Epitopes are typically portions of proteins or peptides, so antibodies typically participate in noncovalent protein-protein interactions.

Time spent:0 sec
Subject:
Biochemistry

QID:402577
System:
Amino Acids and Proteins

The Ankyrin repeat and SOCS box containing protein 4 (Asb-4) is highly expressed in the cytosol of hypothalamus cells during the well-fed state but is down-regulated during fasting. In addition, the protein called Insulin Receptor Substrate 4 (IRS4), known to be involved in appetite regulation, is expressed in the cytosol of the same cells as Asb-4. Accordingly, researchers hypothesized that Asb-4 suppresses appetite through interactions with IRS4.

Small peptides such as myc (EQKLISEEDL) and Flag (DYKDDDDK) are often attached to recombinant proteins to facilitate their purification and detection. To better understand interactions between Asb-4 and IRS4, HEK293 cells were transfected with plasmids encoding IRS4 and different forms of Asb-4, each tagged with either myc or Flag at their N-termini (Flag-IRS4 and myc-Asb-4, respectively). All transfections contained 1 µg of Flag-IRS4 plasmid as well as varying amounts of either myc-Asb-4 plasmid or a plasmid in which a domain called the sb domain of Asb-4 was deleted (myc-Asb-4Δsb). Flag-IRS4 levels from each set of cells were then quantified, as shown in Figure 1.

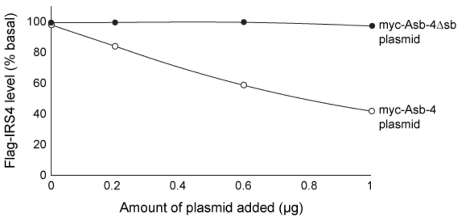


Figure 1 Flag-IRS4 levels in cells with varying amounts of myc-Asb-4 or myc-Asb-4Δsb plasmid added

In a follow-up experiment called coimmunoprecipitation, transfected cells were lysed and the lysates were incubated in columns containing beads linked to α-Flag, an antibody that binds Flag. Proteins that did not bind were washed away. The bound proteins were then visualized by western blot analysis using α-Flag, α-myc, and an antibody against ubiquitin (α-ub). The results are shown in Figure 2. The α-myc antibody detected the presence of myc, indicating that myc-Asb-4 binds to Flag-IRS4.

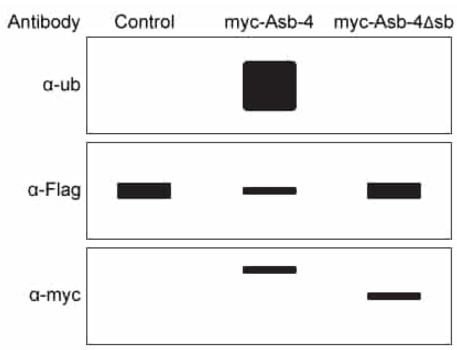


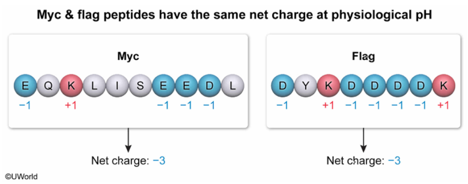
Figure 2 Ubiquitin, Flag-IRS4, and myc-Asb-4 levels in cells transfected with Flag-IRS4 and either an empty vector (control), myc-Asb-4 plasmid, or myc-Asb-4Δsb plasmid

Based on the given sequences of the two peptide tags, which of the following characteristics do the Flag and myc peptides have in common? They have the same:

- A. number of nonpolar residues. [7%]
- B. binding interactions with α-Flag. [24%]
- C. number of acidic residues. [11%]
- ✔ D. net charge at physiological pH. [56%]

Correct
55% answered correctly

Explanation:



The **properties of peptides** and proteins are **determined by** their **amino acid composition**. Each of the 20 canonical amino acids contains a unique side chain that may be polar or nonpolar. Polar amino acids can be further classified as **acidic** (negatively charged at physiological pH), **uncharged**, or **basic** (positively charged). The **net charge** of a peptide or protein is the **sum of the charges** of its **side chains** and the charges of the **N- and C-termini**. At physiological pH, the N-terminus is positively charged and the C-terminus is negatively charged; the two charges cancel, leaving only the side chain charges.

The passage gives the amino acid sequences of both myc (EQKLISEEDL) and Flag (DYKDDDDK). At physiological pH, the myc peptide contains four negatively charged residues: three glutamate residues (E) and one aspartate residue (D). It also contains one positively charged residue (lysine, K). Accordingly, the net charge is −3 (−4 + 1 = −3). The Flag peptide has five negatively charged aspartate residues and two positively charged lysine residues, also yielding a net charge of −3 (−5 + 2 = −3). Therefore, Flag and myc have the **same net charge**.

(Choice A) The myc peptide has three nonpolar (hydrophobic) residues: two leucines (L) and one isoleucine (I). The Flag peptide contains several charged residues and one tyrosine (Y) residue. Most sources classify tyrosine (Y) as polar even though it does also contain a hydrophobic aromatic region. Regardless, myc and Flag do not have the same number of nonpolar residues.

(Choice B) Antibodies are highly specific and typically bind a single chemical group called an epitope. Therefore, an antibody that binds Flag (α-Flag) would most likely *not* bind myc.

(Choice C) The Flag peptide has five acidic residues (aspartate, D), whereas myc has only four (three glutamate, E, and one aspartate, D).

Educational objective:
Different peptides and proteins have different properties based on their amino acid composition. At any given pH, the net charge of a peptide is the sum of the charges of its side chains and the N- and C-termini. At physiological pH, the positive charge of the N-terminus cancels the negative charge of the C-terminus.

Time spent:0 sec

QID:402578

Subject:
Biochemistry

System:
Amino Acids and Proteins

The Ankyrin repeat and SOCS box containing protein 4 (Asb-4) is highly expressed in the cytosol of hypothalamus cells during the well-fed state but is down-regulated during fasting. In addition, the protein called Insulin Receptor Substrate 4 (IRS4), known to be involved in appetite regulation, is expressed in the cytosol of the same cells as Asb-4. Accordingly, researchers hypothesized that Asb-4 suppresses appetite through interactions with IRS4.

Small peptides such as myc (EQKLISEEDL) and Flag (DYKDDDDK) are often attached to recombinant proteins to facilitate their purification and detection. To better understand interactions between Asb-4 and IRS4, HEK293 cells were transfected with plasmids encoding IRS4 and different forms of Asb-4, each tagged with either myc or Flag at their N-termini (Flag-IRS4 and myc-Asb-4, respectively). All transfections contained 1 µg of Flag-IRS4 plasmid as well as varying amounts of either myc-Asb-4 plasmid or a plasmid in which a domain called the sb domain of Asb-4 was deleted (myc-Asb-4Δsb). Flag-IRS4 levels from each set of cells were then quantified, as shown in Figure 1.

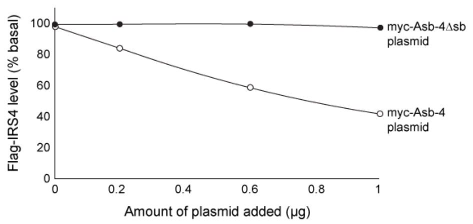


Figure 1 Flag-IRS4 levels in cells with varying amounts of myc-Asb-4 or myc-Asb-4Δsb plasmid added

In a follow-up experiment called coimmunoprecipitation, transfected cells were lysed and the lysates were incubated in columns containing beads linked to α-Flag, an antibody that binds Flag. Proteins that did not bind were washed away. The bound proteins were then visualized by western blot analysis using α-Flag, α-myc, and an antibody against ubiquitin (α-ub). The results are shown in Figure 2. The α-myc antibody detected the presence of myc, indicating that myc-Asb-4 binds to Flag-IRS4.

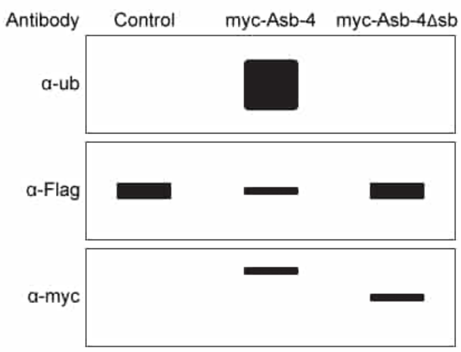


Figure 2 Ubiquitin, Flag-IRS4, and myc-Asb-4 levels in cells transfected with Flag-IRS4 and either an empty vector (control), myc-Asb-4 plasmid, or myc-Asb-4Δsb plasmid

To conclude that IRS4 binds Asb-4, what assumption must be made about the recombinant proteins?

A. Recombinant Flag-IRS4 and myc-Asb-4 are expressed at similar levels. [23%]

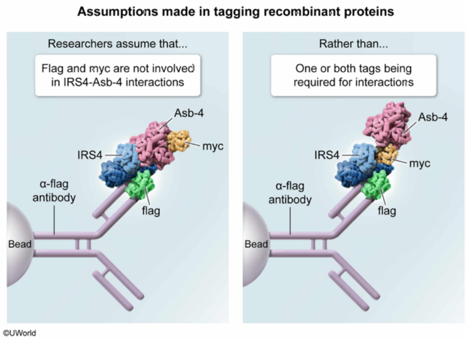
B. HEK293 cells do not express IRS4 and Asb-4 endogenously. [18%]

C. Flag and myc are not involved in interactions between Flag-IRS4 and myc-Asb-4. [43%]

D. Flag-IRS4 and myc-Asb-4 are tagged at their N-termini only. [14%]

Correct
42% answered correctly

Explanation:



The **function of a protein** is **determined by the sequence of amino acids** that make up that protein, known as the **primary structure**. The primary structure dictates how the protein folds into secondary and tertiary structures, which are required for protein function. Therefore, **altering the amino acid sequence can affect protein function**.

According to the passage, small peptides (Flag or myc) were added to the N-termini of both IRS4 and Asb-4, which altered their primary structures. In the experiment described, cell lysates containing these proteins were run on a column with beads containing the anti-Flag antibody (α-Flag). Unbound proteins were then washed away. The retained proteins underwent western blot analysis and were visualized using various antibodies. These retained proteins included the Flag-tagged proteins that bound the column directly, as well as other proteins that bound *indirectly* through interactions with the Flag-tagged proteins.

As expected, the Flag tag was detected. The myc tag was also detected, meaning myc-Asb-4 was retained on the column. Because myc cannot bind the α-Flag antibodies on the column, the only way it can be retained is if myc-Asb-4 binds to the Flag-IRS4 proteins that *did* bind to the column. This binding is *most likely* due to interactions between the Asb-4 and IRS4 proteins themselves.

However, it is possible that the binding interaction involved one or both of the tags. For example, the myc tag itself might directly bind IRS4. In such a case, the conclusion that Asb-4 binds to IRS4 would be invalid. Therefore, to conclude that Asb-4 binds IRS4, the **researchers must assume** that **Flag and myc** are **not involved in the binding**. This assumption could be confirmed by repeating the experiment in the absence of tags or using different tags.

(Choice A) Protein binding interactions may occur regardless of whether the proteins of interest are expressed at similar levels, so no assumptions need to be made about expression levels.

(Choice B) Binding interactions between the recombinant proteins can occur regardless of whether HEK293 cells express the proteins endogenously.

(Choice D) The passage states that the Flag and myc tags were placed at the N-termini of each protein, so no assumptions need to be made about the tag positions.

Educational objective:
The biological function of a protein is determined by the sequence of amino acids that make up that protein (primary structure). Altering the sequence of a protein may alter its function.

Time spent:0 sec
Subject:
Biochemistry

QID:402579
System:
Amino Acids and Proteins

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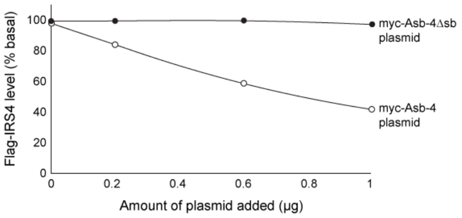


Figure 1 Flag-IRS4 levels in cells with varying amounts of myc-Asb-4 or myc-Asb-4Δsb plasmid added

In a follow-up experiment called coimmunoprecipitation, transfected cells were lysed and the lysates were incubated in columns containing beads linked to α-Flag, an antibody that binds Flag. Proteins that did not bind were washed away. The bound proteins were then visualized by western blot analysis using α-Flag, α-myc, and an antibody against ubiquitin (α-ub). The results are shown in Figure 2. The α-myc antibody detected the presence of myc, indicating that myc-Asb-4 binds to Flag-IRS4.

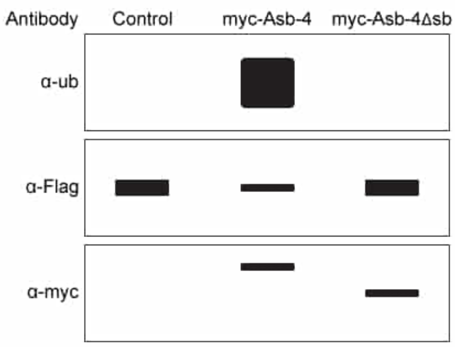


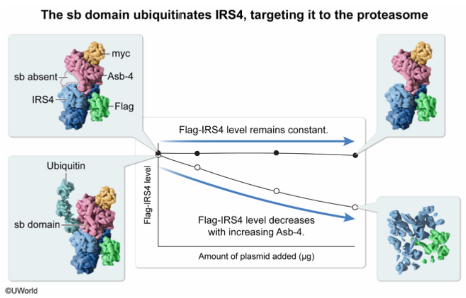
Figure 2 Ubiquitin, Flag-IRS4, and myc-Asb-4 levels in cells transfected with Flag-IRS4 and either an empty vector (control), myc-Asb-4 plasmid, or myc-Asb-4Δsb plasmid

Which of the following is the most likely mechanism by which the sb region of Asb-4 decreases IRS4 levels?

- A. The sb domain facilitates ubiquitination of IRS4, targeting it to the proteasome. [62%]
- B. The sb domain induces IRS4 expression by entering the nucleus to bind DNA. [6%]
- C. The sb domain prevents degradation of IRS4 by inhibiting proteases. [12%]
- D. The sb domain becomes ubiquitinated, causing IRS4 to enter the lysosome. [18%]

Correct
61% answered correctly

Explanation:



Protein levels within a cell can be controlled by both the rate of **protein synthesis** (expression) and the rate of **protein degradation**. Protein expression is controlled by several mechanisms, including transcription factors and RNA processing. Degradation is typically induced by sending proteins to the lysosome or the proteasome.

Several pieces of evidence indicate the mechanism by which the sb domain of Asb-4 regulates IRS4 levels. First, Figure 1 shows that the level of Flag-IRS4 *decreases* in proportion to the level of myc-Asb-4 expression. Therefore, Asb-4 causes a *decrease* in IRS4 levels, either by inhibiting expression or by inducing degradation, or both. When the sb domain is removed (myc-Asb-4Δsb), IRS4 levels are constant, indicating that the sb domain of Asb-4 is *required* for regulation of IRS4 levels.

Second, Figure 2 shows that when myc-Asb-4 is present, a ubiquitin tag is detected. Because this experiment was performed by binding proteins to beads containing the α-Flag antibody, all proteins other than Flag-IRS4 were washed away or bound to beads *indirectly*, most likely by interacting with IRS4. Figure 2 shows that ubiquitin binds to IRS4 only when the sb domain of myc-Asb-4 is present. Therefore, Asb-4 likely causes ubiquitination of Flag-IRS4. Ubiquitin is *not* detected when the sb domain is removed, indicating that this domain is *required* for ubiquitination. **Proteins with ubiquitin tags** are typically **targeted to the proteasome** for destruction.

Based on all the data together, the sb domain of Asb-4 most likely regulates IRS4 levels by targeting IRS4 to the proteasome for destruction.

(Choices B and C) If the sb domain *induced* IRS4 expression or *inhibited* IRS4 proteolysis (degradation), IRS4 levels would be expected to *increase*, not decrease, when the sb domain is present.

(Choice D) Ubiquitin tags send proteins to the *proteasome*, not the lysosome, for destruction. The lysosome typically degrades secretory proteins, whereas the passage states that IRS4 is cytosolic.

Educational objective:

Protein levels in a cell are controlled by expression and degradation rates. Expression is controlled by mechanisms such as transcription factors and RNA processing, whereas degradation is controlled by targeting secretory proteins to the lysosome and cytosolic proteins to the proteasome. Ubiquitin tags mark cytosolic proteins for proteasomal destruction.

Time spent:0 sec
Subject:
Biochemistry

QID:402580
System:
Amino Acids and Proteins